

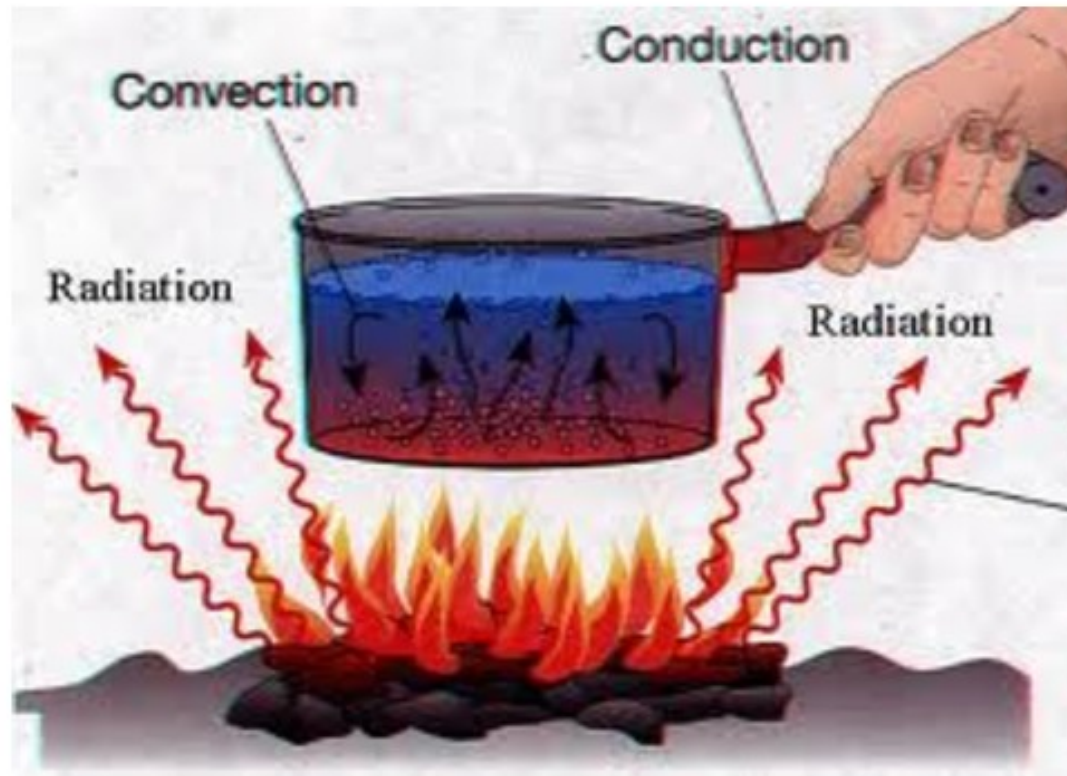
The Transfer of Heat

Chapter 13



The Transfer of Heat

3 WAYS for heat to travel :

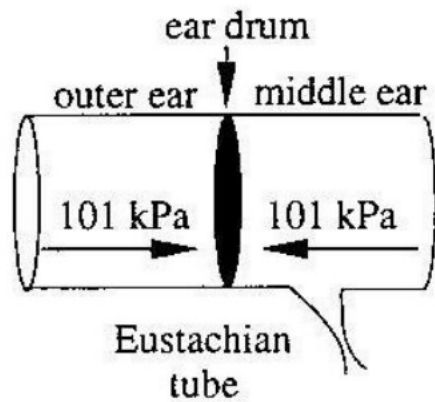


13.1 Convection

CONVECTION: a way to transfer heat

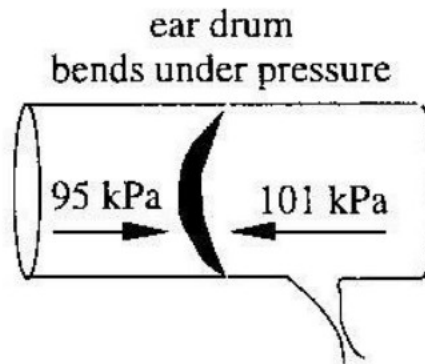
Convection is the process in which heat is carried from one place to another by the bulk movement of a fluid.





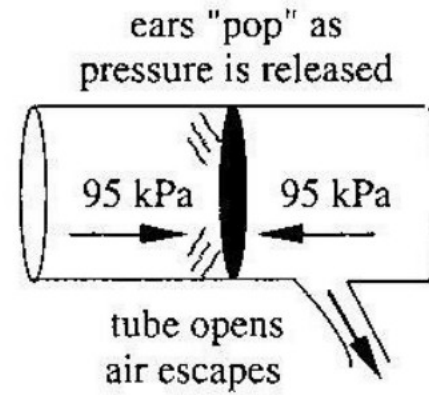
SEA LEVEL

B



MOUNTAINS

C



MOUNTAINS

D

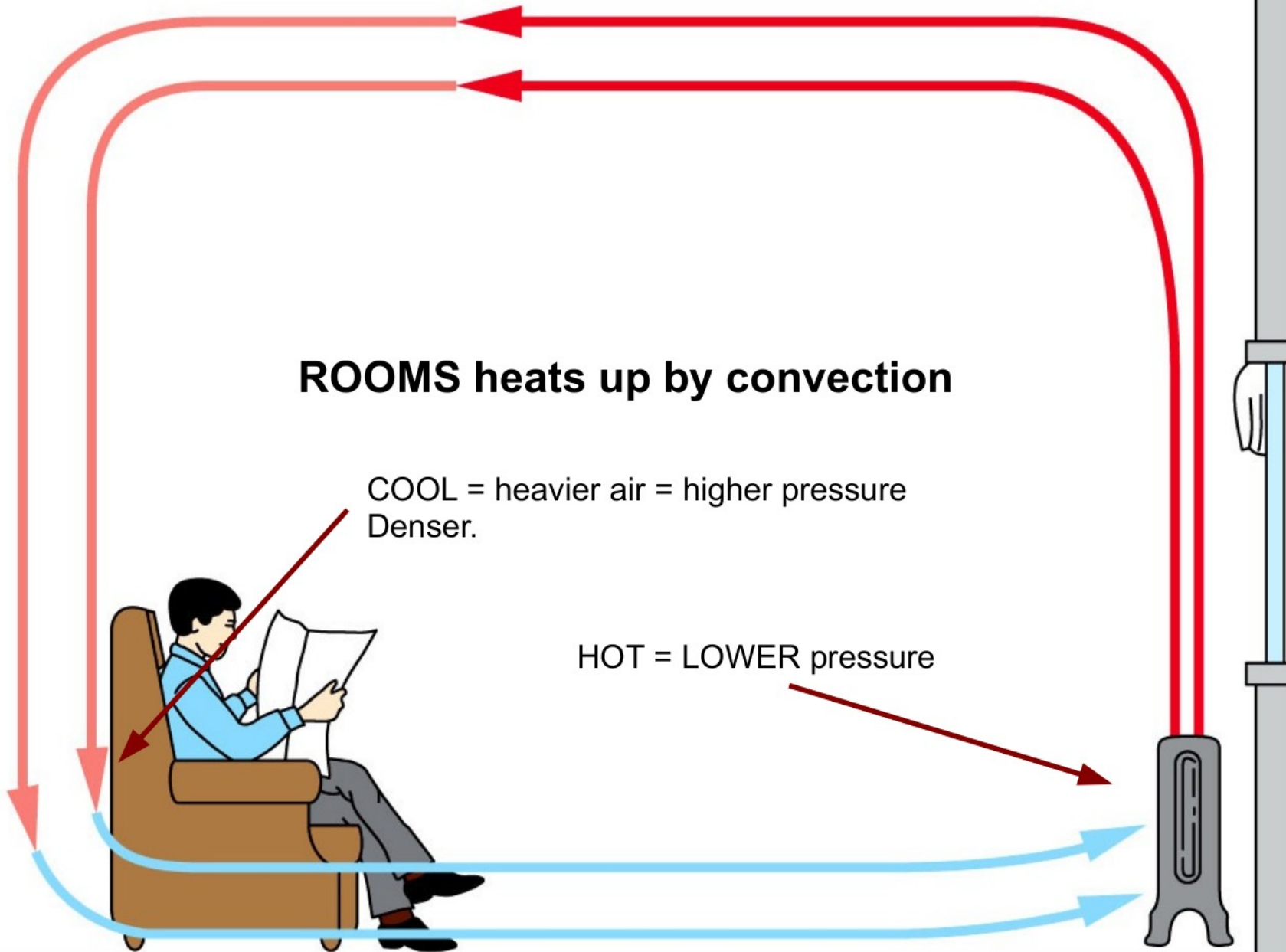


Air moves from high pressure to low pressure

ROOMS heats up by convection

COOL = heavier air = higher pressure
Denser.

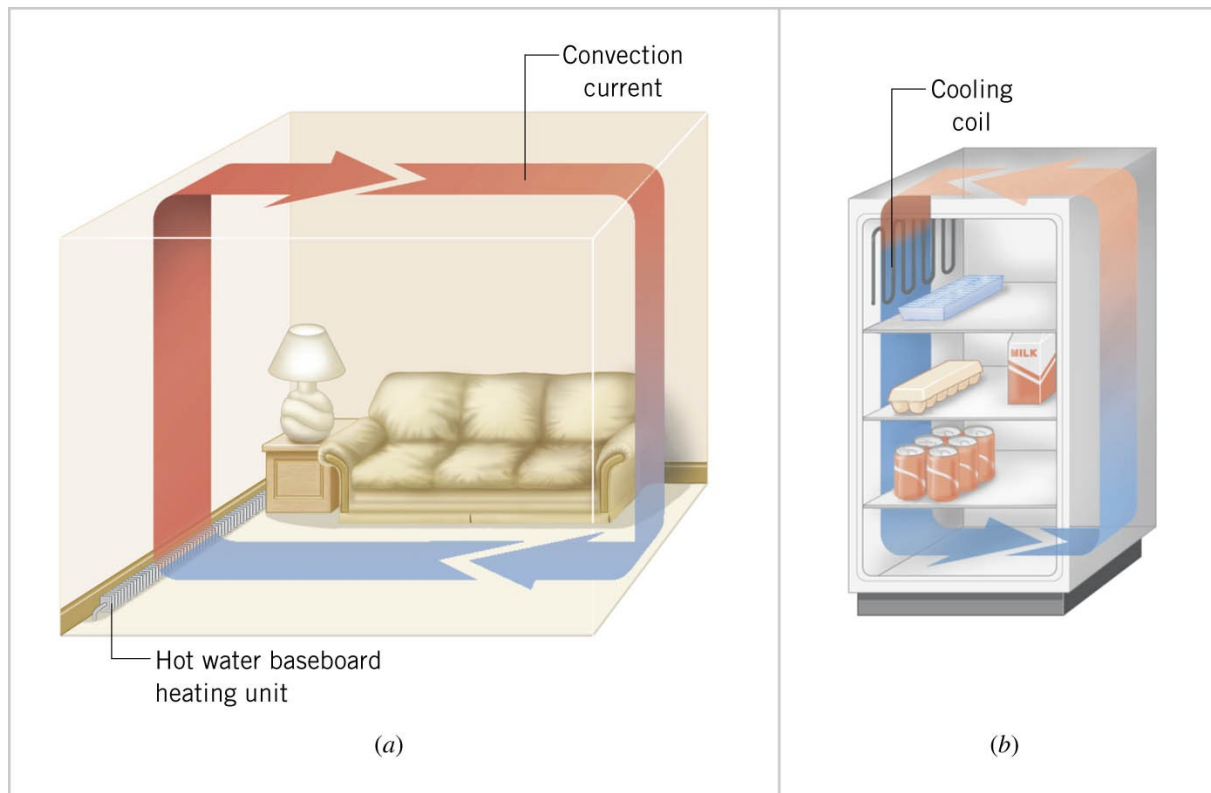
HOT = LOWER pressure



13.1 Convection

Conceptual Example 1 Hot Water Baseboard Heating and Refrigerators

Hot water baseboard heating units are mounted on the wall next to the floor. The cooling coil in a refrigerator is mounted near the top of the refrigerator. Each location is designed to maximize the production of convection currents. Explain how.



From your book p. 386
Thermals used by gliders or hawks.

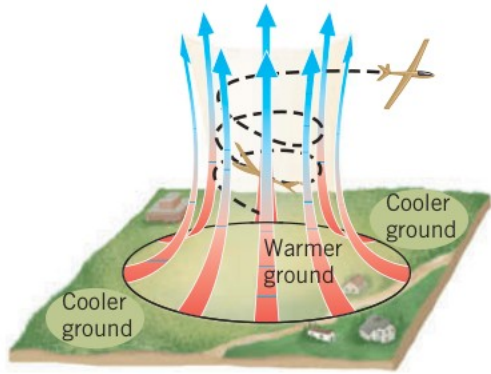
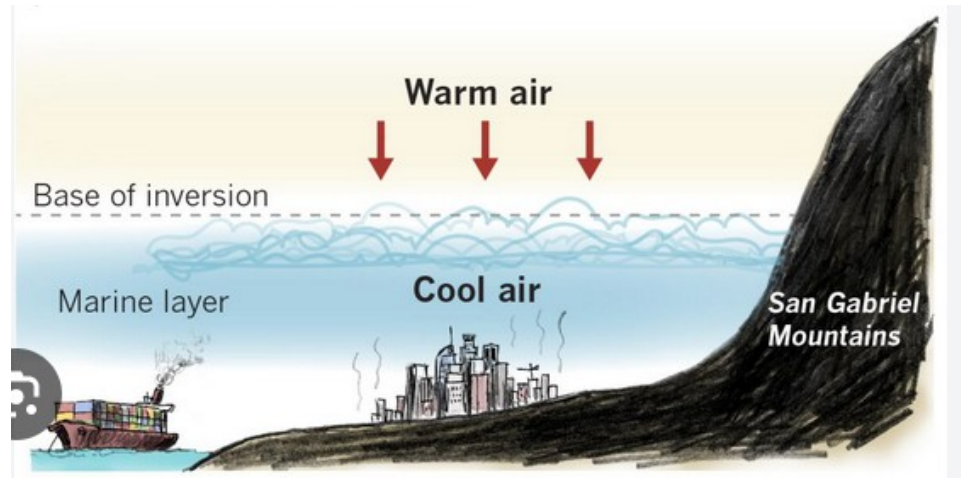


Figure 13.4 Updrafts, or thermals, are caused by the convective movement of air that the ground has warmed.

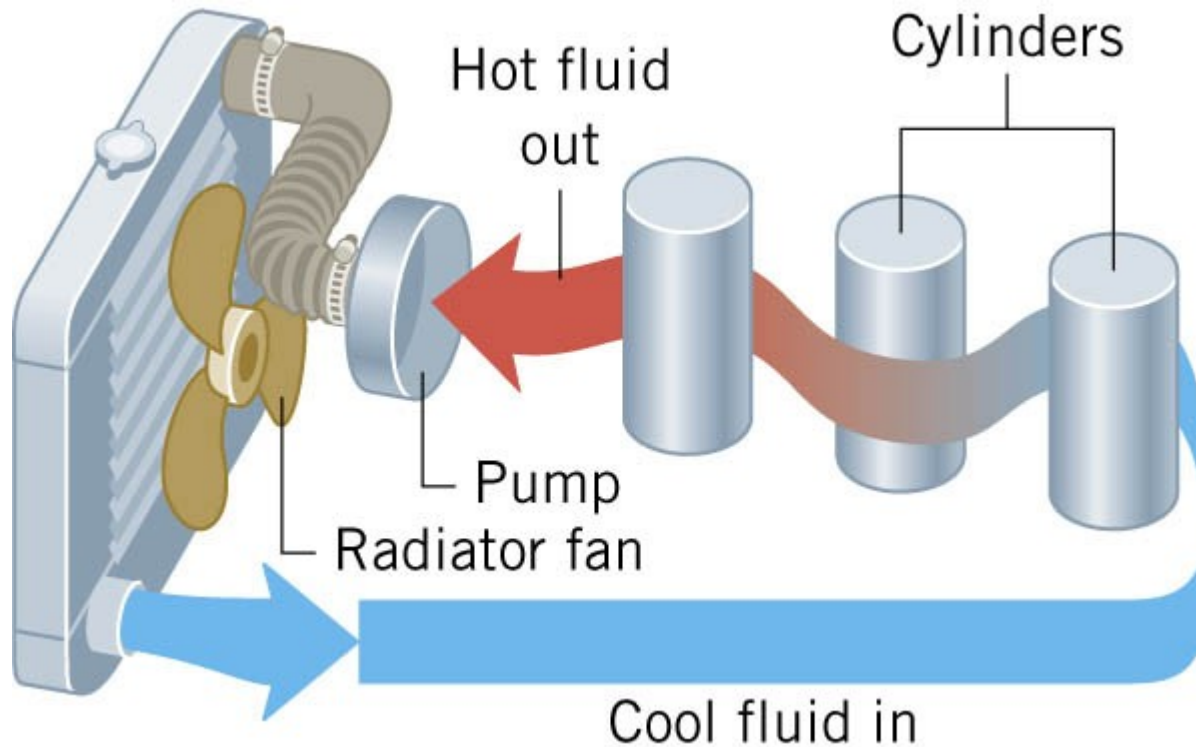


inversion



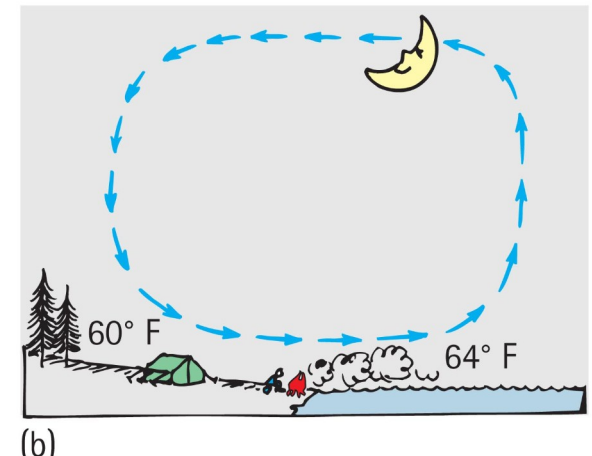
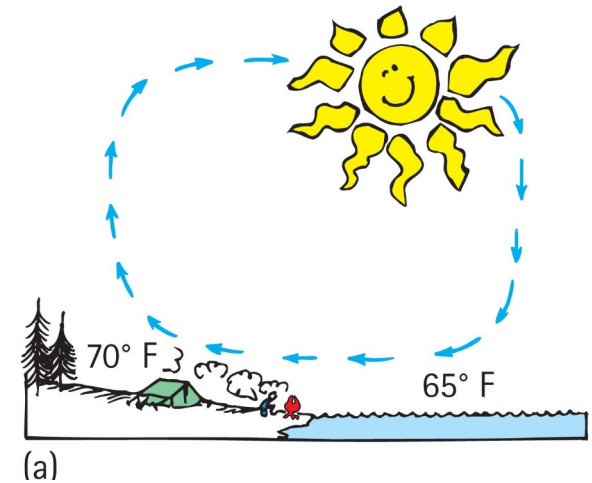
13.1 Convection
p. 386 of your textbook

Forced Convection
Also in a computer.



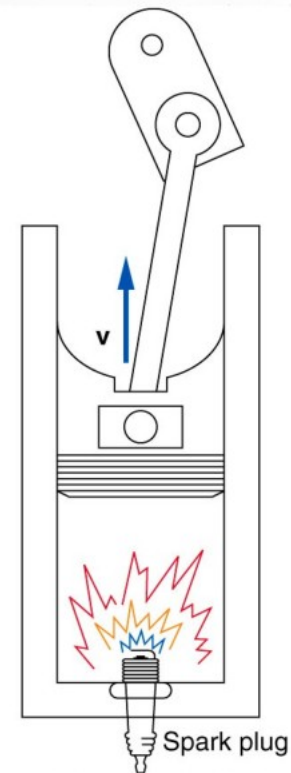
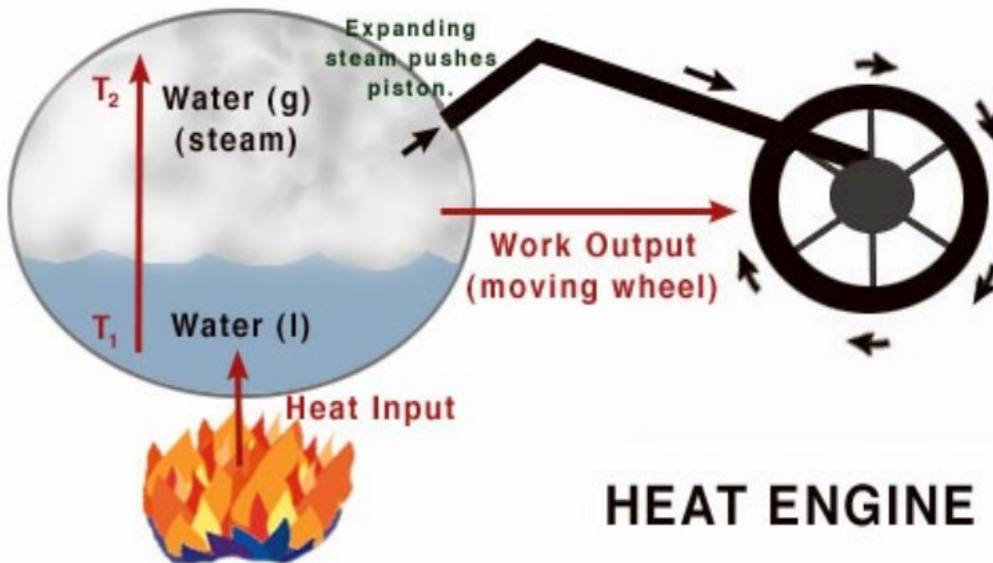
Convection, Continued-2

- Winds
 - Result of uneven heating of the air near the ground
 - Absorption of Sun's energy occurs more readily on different parts of Earth's surface.
 - Sea breeze
 - The ground warms more than water in the daytime.
 - Warm air close to the ground rises and is replaced by cooler air from above the water.



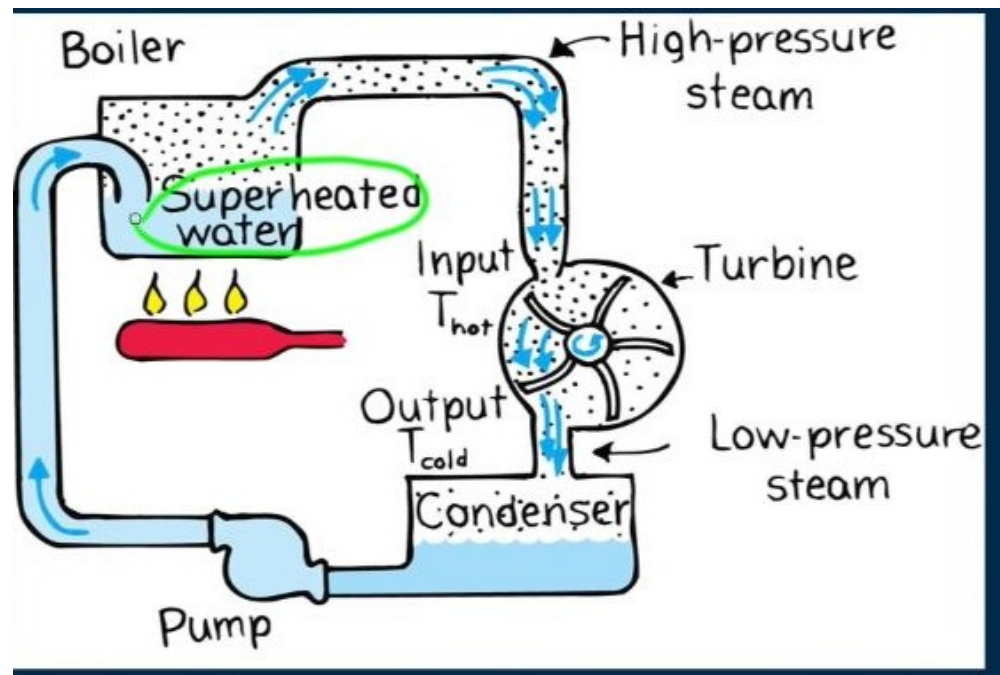
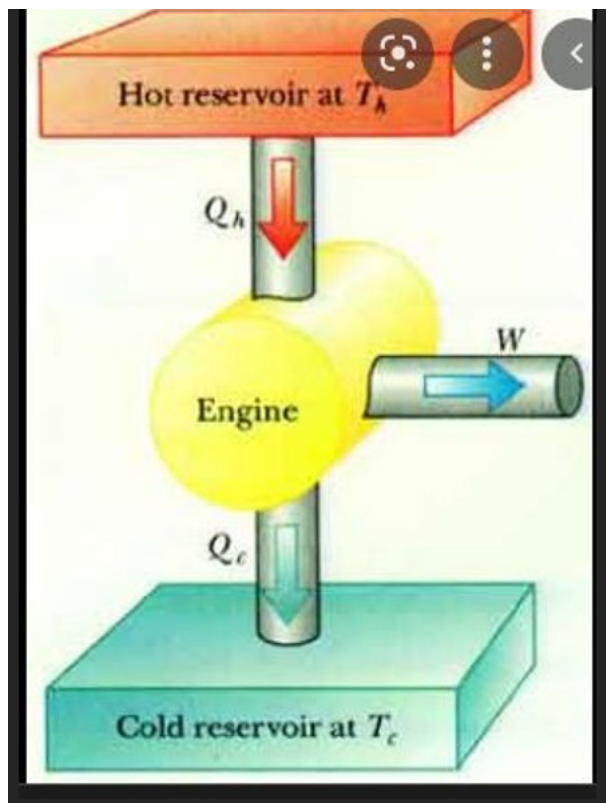
Heat engine → Industrial revolution !

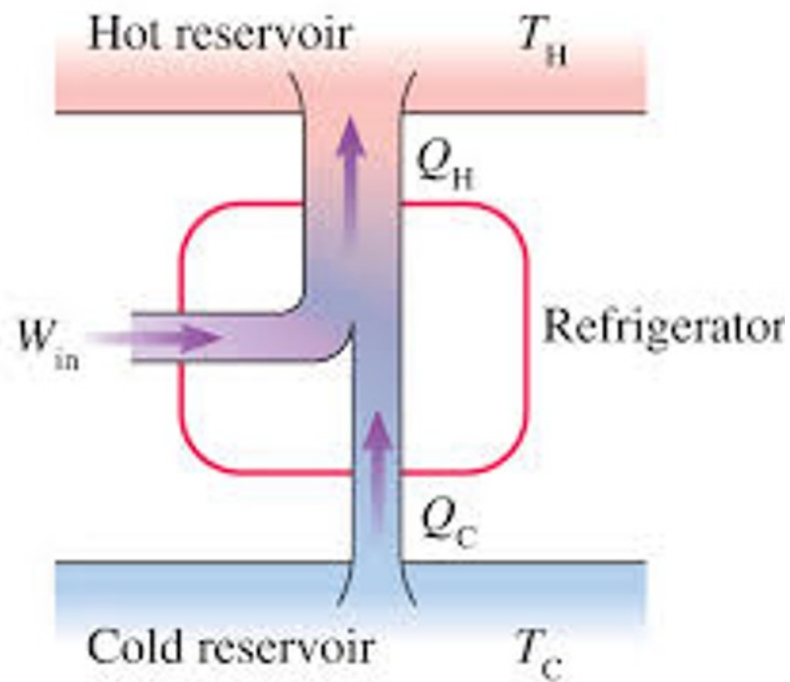
See chapter: heat and atoms
Physics behind the headlines
Prof. Muller



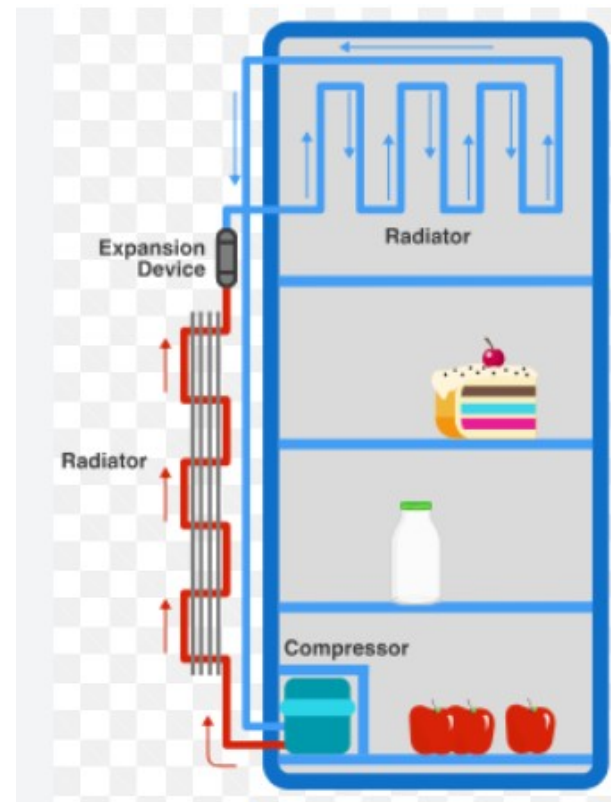
Higher the difference
In temperature →
Higher the efficiency.

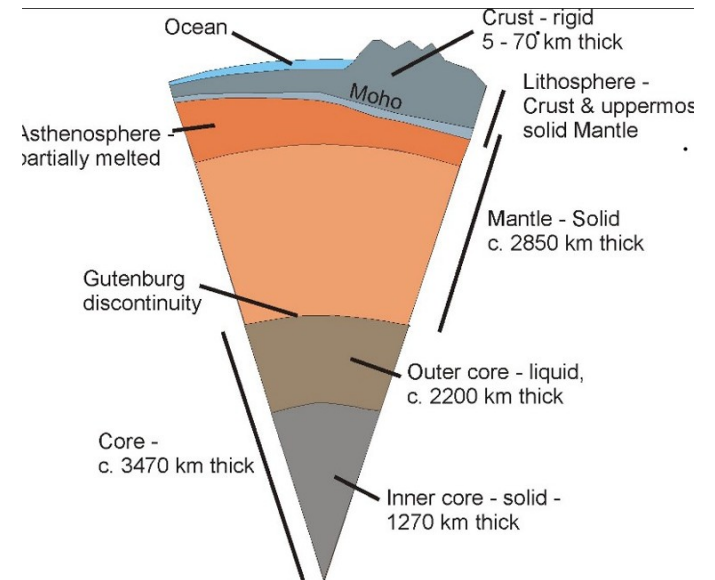
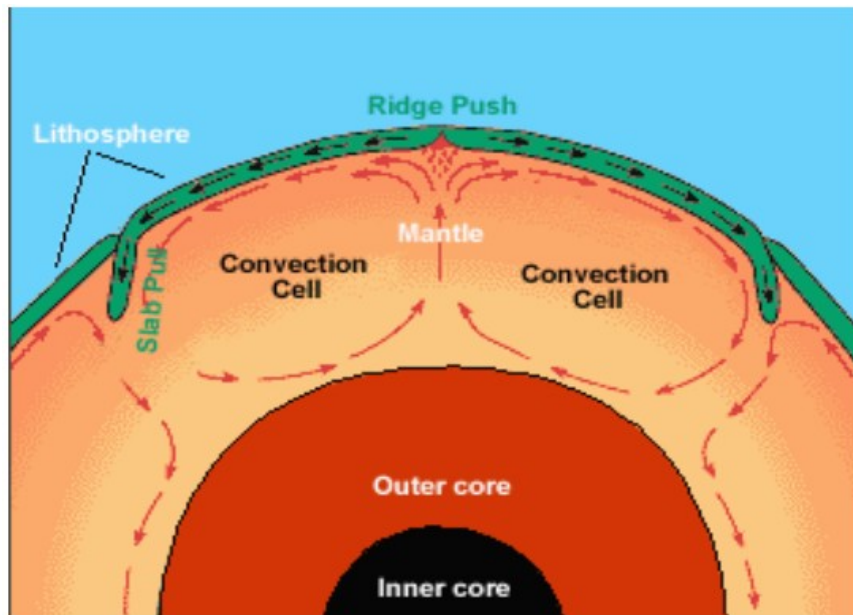
The beetle was very powerful
But too much CO emitted





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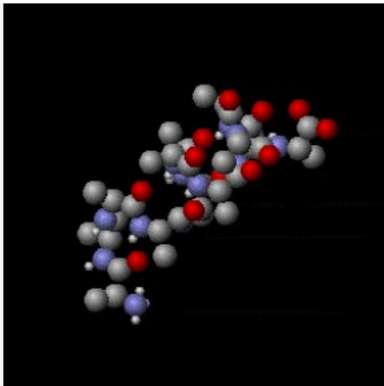
CONDUCTION

Conduction is the process whereby heat is transferred directly through a material, with any bulk motion of the material playing no role in the transfer.

One mechanism for conduction occurs when the atoms or molecules in a hotter part of the material vibrate or move with greater energy than those in a cooler part.

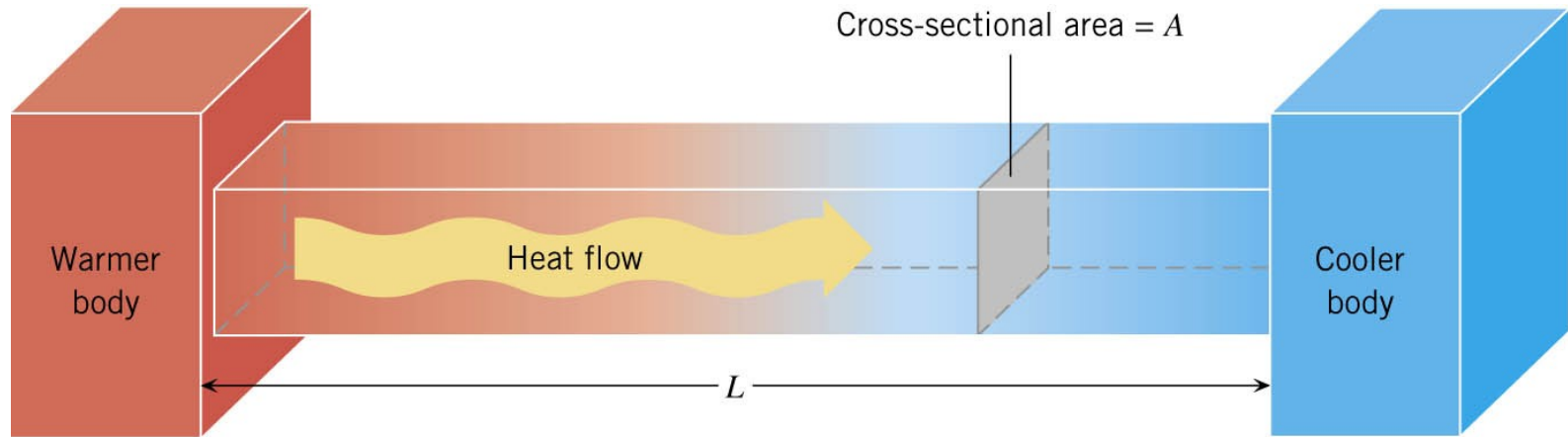
By means of collisions, the more energetic molecules pass on some of their energy to their less energetic neighbors.

Materials that conduct heat well are called ***thermal conductors***, and those that conduct heat poorly are called ***thermal insulators***.



Molecules , particles random motion

13.2 Conduction



The amount of heat Q that is conducted through the bar depends on a number of factors:

1. The time during which conduction takes place.
2. The temperature difference between the ends of the bar.
3. The cross sectional area of the bar.
4. The length of the bar.

13.2 Conduction

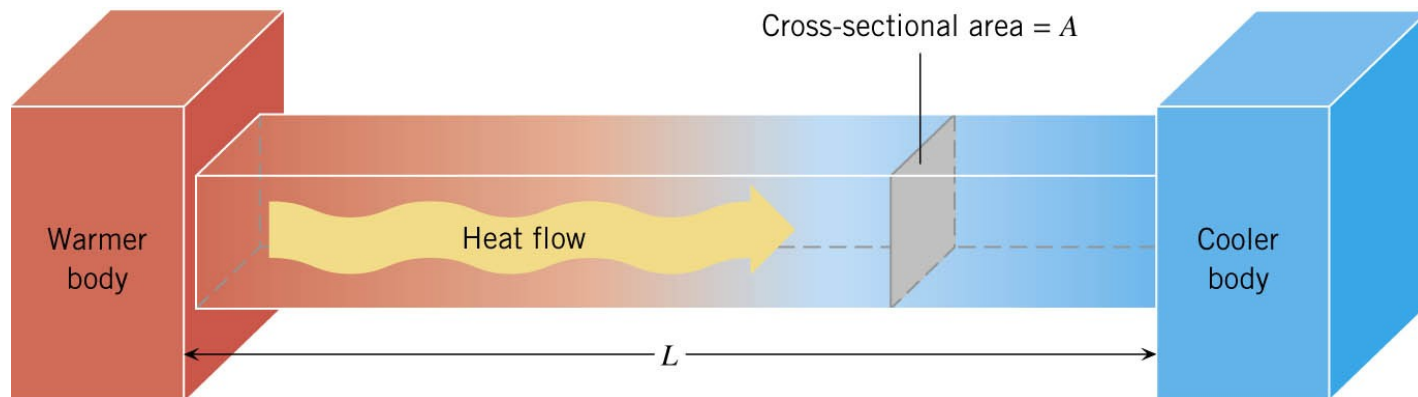
CONDUCTION OF HEAT THROUGH A MATERIAL

The heat Q conducted during a time t through a bar of length L and cross-sectional area A is

$$Q = \frac{(kA \Delta T) t}{L}$$

thermal conductivity

SI Units of Thermal Conductivity: $\text{J}/(\text{s} \cdot \text{m} \cdot \text{C}^\circ)$



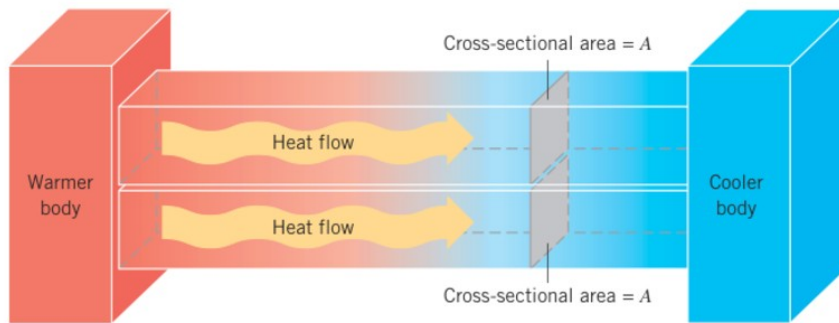
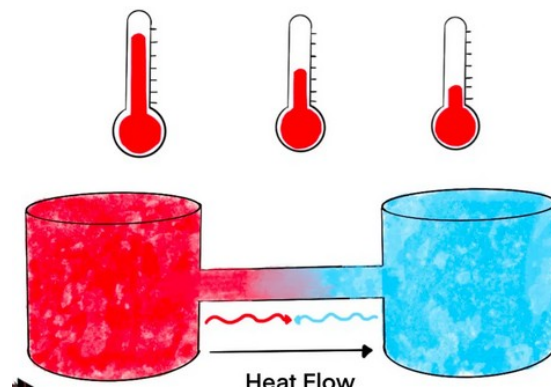
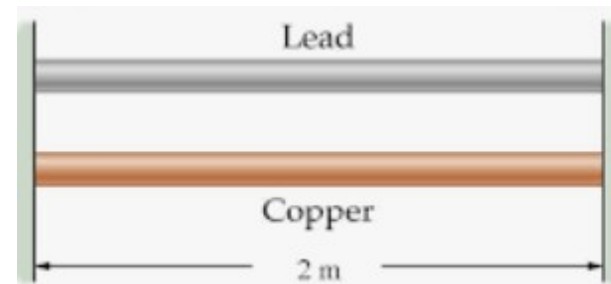
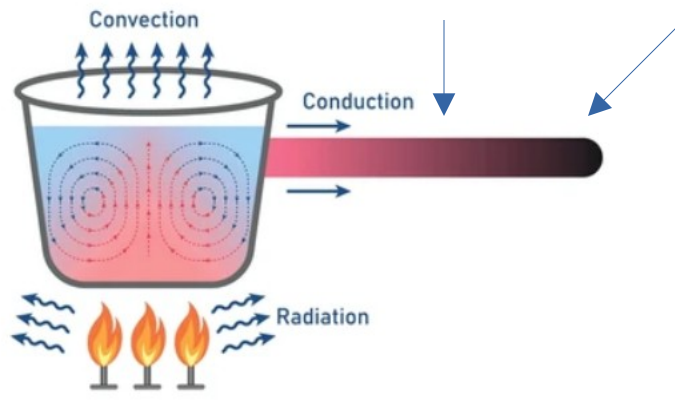


Figure 13.8 Twice as much heat flows through two identical bars as through one.



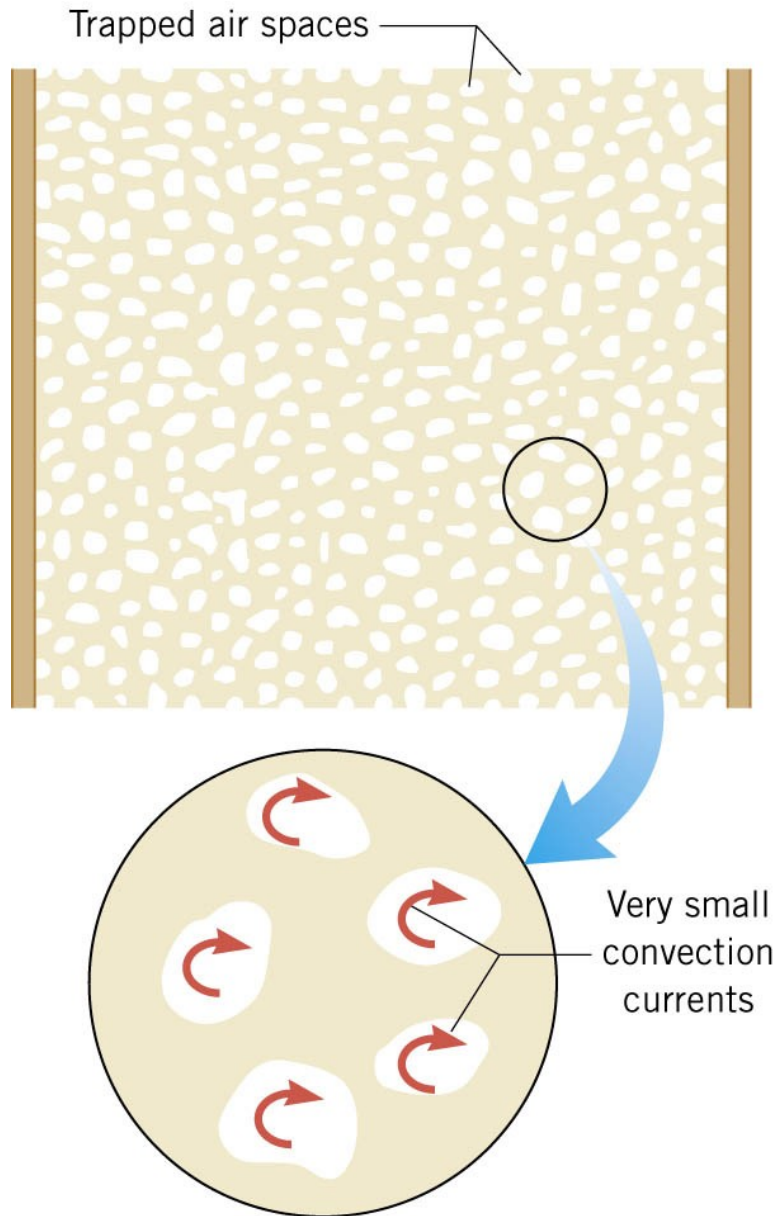
13.2 Conduction

**Table 13.1 Thermal Conductivities^a
of Selected Materials**

Substance	Thermal Conductivity, k [J/(s · m · °C)]
Metals	
Aluminum	240
Brass	110
Copper	390
Iron	79
Lead	35
Silver	420
Steel (stainless)	14
Gases	
Air	0.0256
Hydrogen (H ₂)	0.180
Nitrogen (N ₂)	0.0258
Oxygen (O ₂)	0.0265
Other Materials	
Asbestos	0.090
Body fat	0.20
Concrete	1.1
Diamond	2450
Glass	0.80
Goose down	0.025
Ice (0 °C)	2.2
Styrofoam	0.010
Water	0.60
Wood (oak)	0.15
Wool	0.040

^a Except as noted, the values pertain to temperatures near 20 °C.

13.2 Conduction



Materials with dead air spaces are usually excellent thermal insulators.

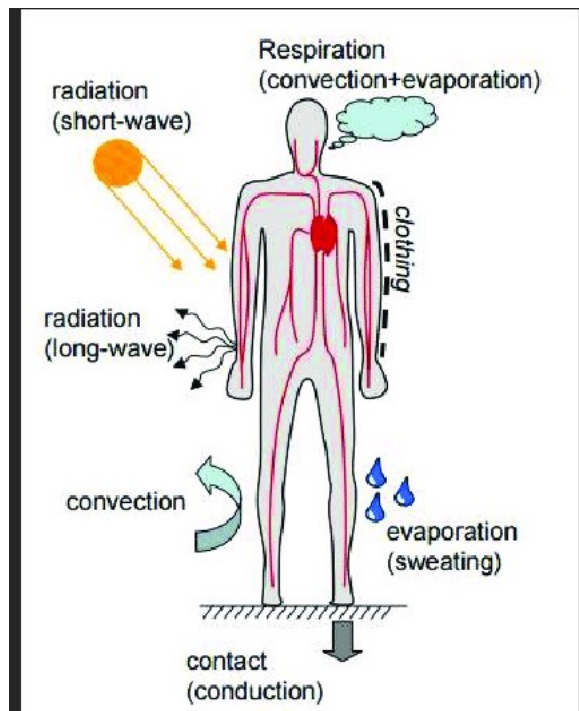


The Transfer of Heat

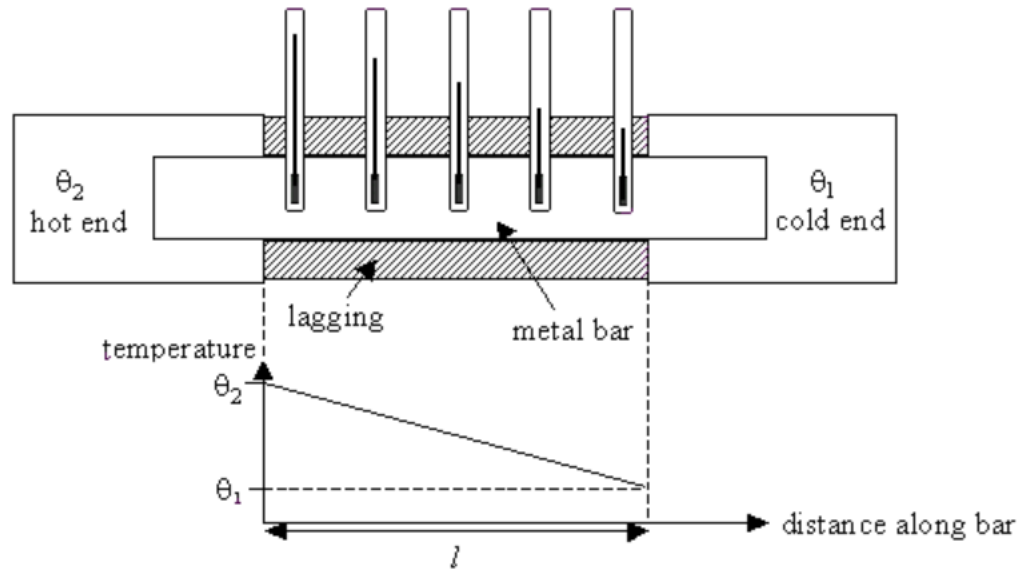
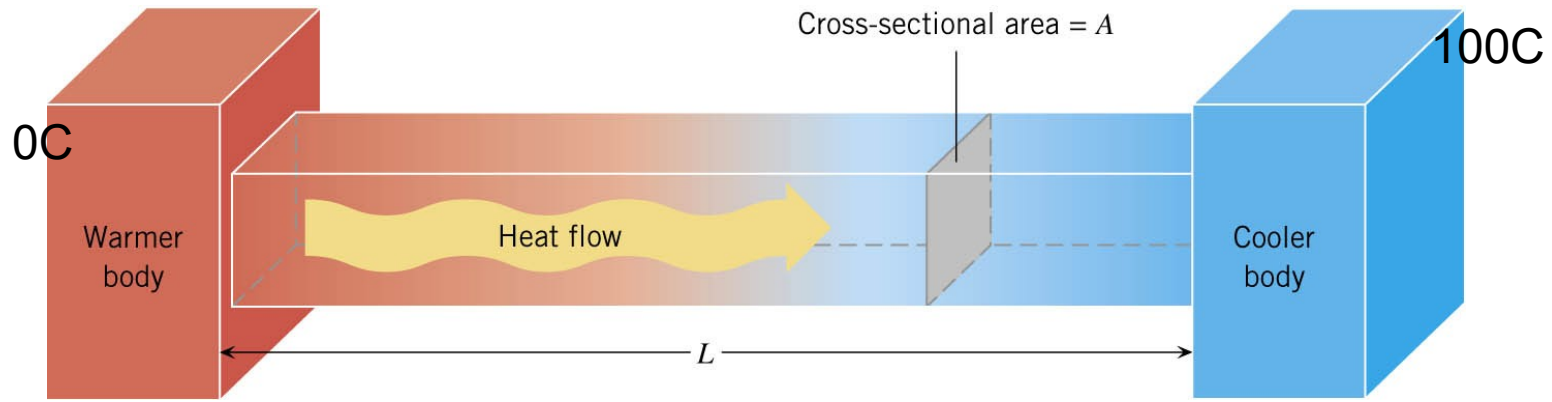
Example 2 The Physics of Heat Transfer in the Human Body

When excessive heat is produced within the body, it must be transferred to the skin and dispersed if the temperature at the body interior is to be maintained at the normal value of $37.0\text{ }^{\circ}\text{C}$. One possible mechanism for transfer is conduction through body fat. Suppose that heat travels through 0.030 m of fat in reaching the skin, which has a total surface area of 1.7 m^2 and a temperature of $34.0\text{ }^{\circ}\text{C}$. Find the amount of heat that reaches the skin in half an hour (1800 s).

Reasoning and Solution In Table 13.1 the thermal conductivity of body fat is given as $k = 0.20\text{ J}/(\text{s} \cdot \text{m} \cdot \text{C}^{\circ})$. According to Equation 13.1,



p. 389

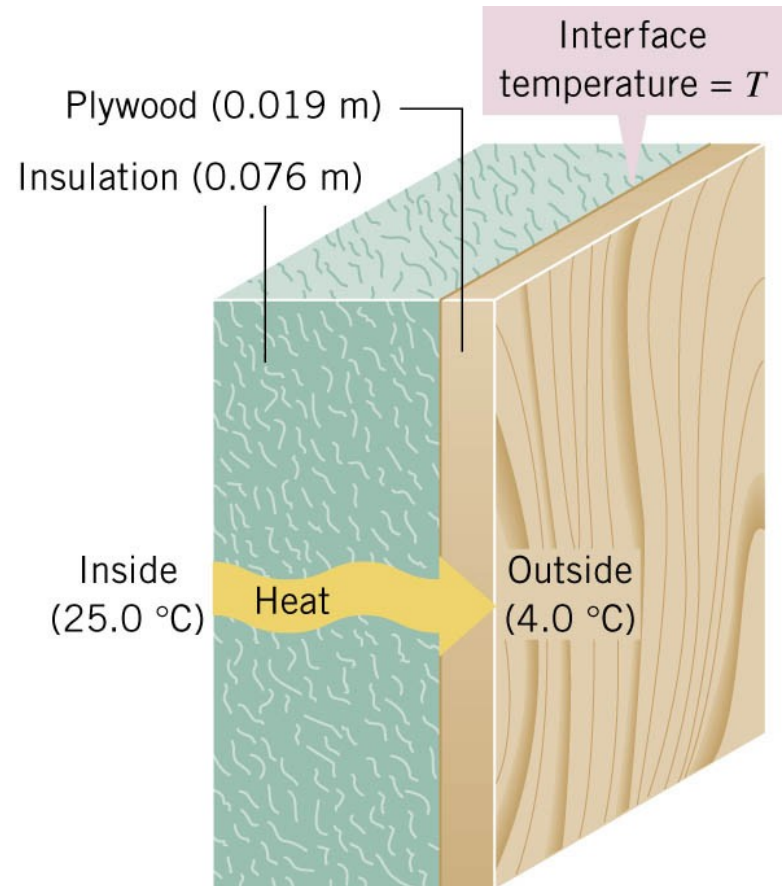


13.2 Conduction

Example 4 Layered insulation

One wall of a house consists of plywood backed by insulation. The thermal conductivities of the insulation and plywood are, respectively, 0.030 and $0.080 \text{ J}/(\text{s}\cdot\text{m}\cdot\text{C}^\circ)$, and the area of the wall is 35m^2 .

Find the amount of heat conducted through the wall in one hour.

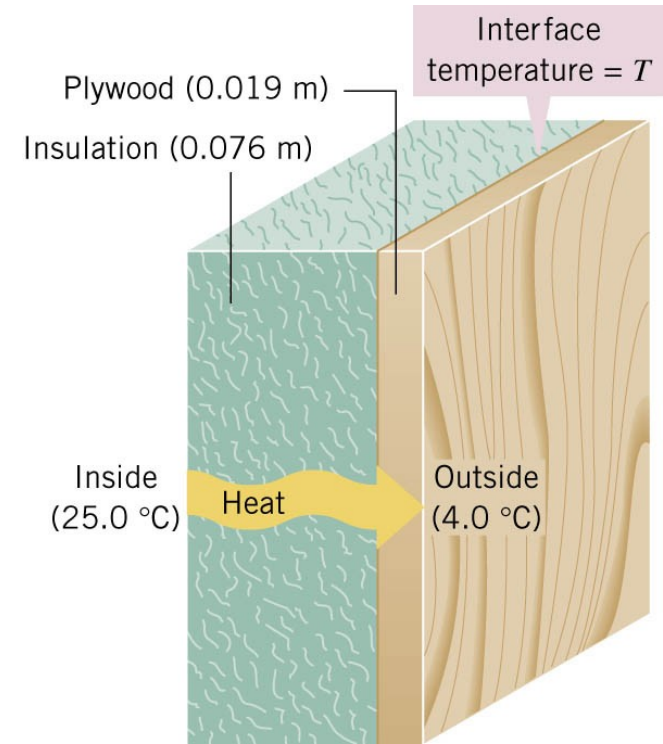


13.2 Conduction

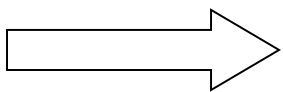
$$Q = Q_{\text{insulation}} = Q_{\text{plywood}}$$

But first we must solve for the interface temperature.

$$\left[\frac{(kA \Delta T) t}{L} \right]_{\text{insulation}} = \left[\frac{(kA \Delta T) t}{L} \right]_{\text{plywood}}$$

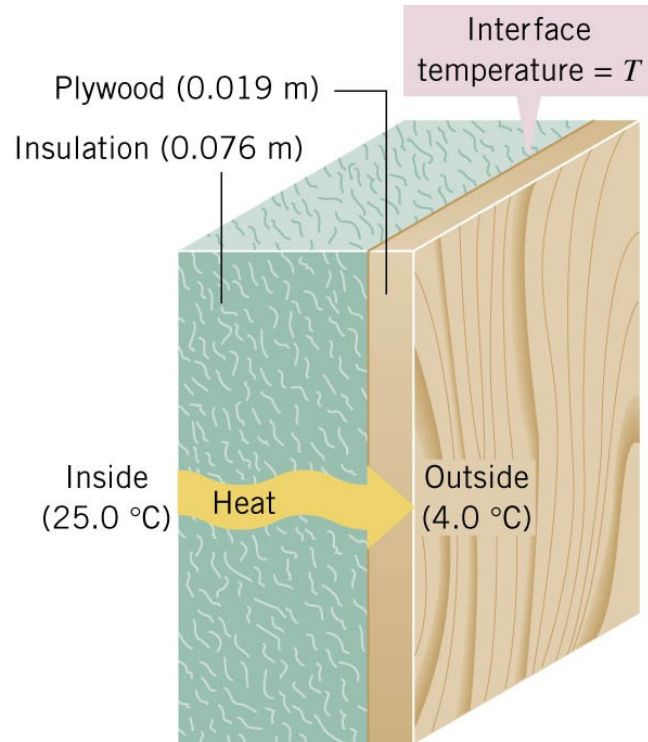


$$\frac{[0.030 \text{ J}/(\text{s} \cdot \text{m} \cdot \text{C}^\circ)] A (25.0^\circ \text{C} - T) t}{0.076 \text{ m}} = \frac{[0.080 \text{ J}/(\text{s} \cdot \text{m} \cdot \text{C}^\circ)] A (T - 4.0^\circ \text{C}) t}{0.019 \text{ m}}$$



$$T = 5.8^\circ \text{C}$$

13.2 Conduction

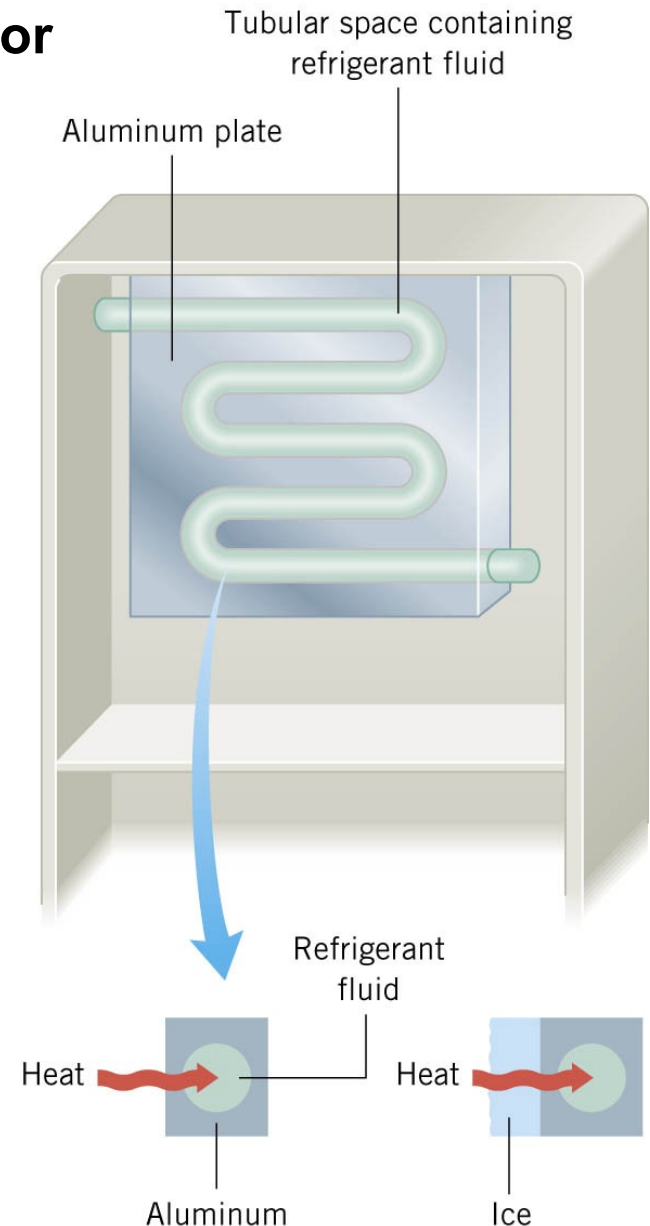


$$Q_{\text{insulation}} = \frac{[0.030 \text{ J}/(\text{s} \cdot \text{m} \cdot \text{C}^\circ)](35 \text{ m}^2)(25.0^\circ \text{C} - 5.8^\circ \text{C})(3600 \text{ s})}{0.076 \text{ m}}$$

$$9.5 \times 10^5 \text{ J}$$

Conceptual Example 5 An Iced-Up Refrigerator

In a refrigerator, heat is removed by a cold refrigerant fluid that circulates within a tubular space embedded within a metal plate. Decide whether the plate should be made from aluminum or stainless steel and whether the arrangement works better or worse when it becomes coated with a layer of ice.



From textbook



Figure 13.11 When the temperature dips below freezing, strawberry growers spray their plants with water to put a coat of ice on them. Because of the heat released when the water freezes and because of the relatively small thermal conductivity of ice, this procedure protects the plants against subfreezing temperatures. The photograph shows berries that survived a temperature of -3°C with little or no damage. (Pierre Ducharme/Reuters/Landov LLC)

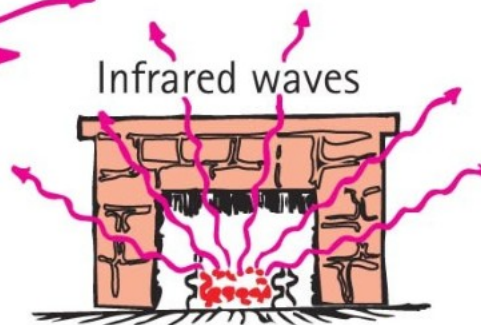
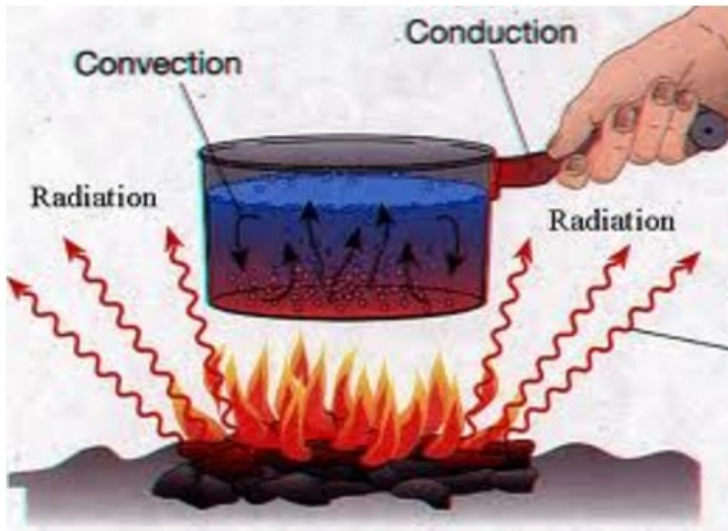
RADIATION

Radiation is the process in which energy is transferred by means of electromagnetic waves.

EM waves carry energy. UV from the Sun
→ burn

Radiant energy is the energy of electromagnetic waves.

This form of energy can travel through space and does not require a medium to propagate.



<https://phet.colorado.edu/sims/cheerpj/microwaves/latest/microwaves.html?simulation=microwaves>

Go back to slide 20

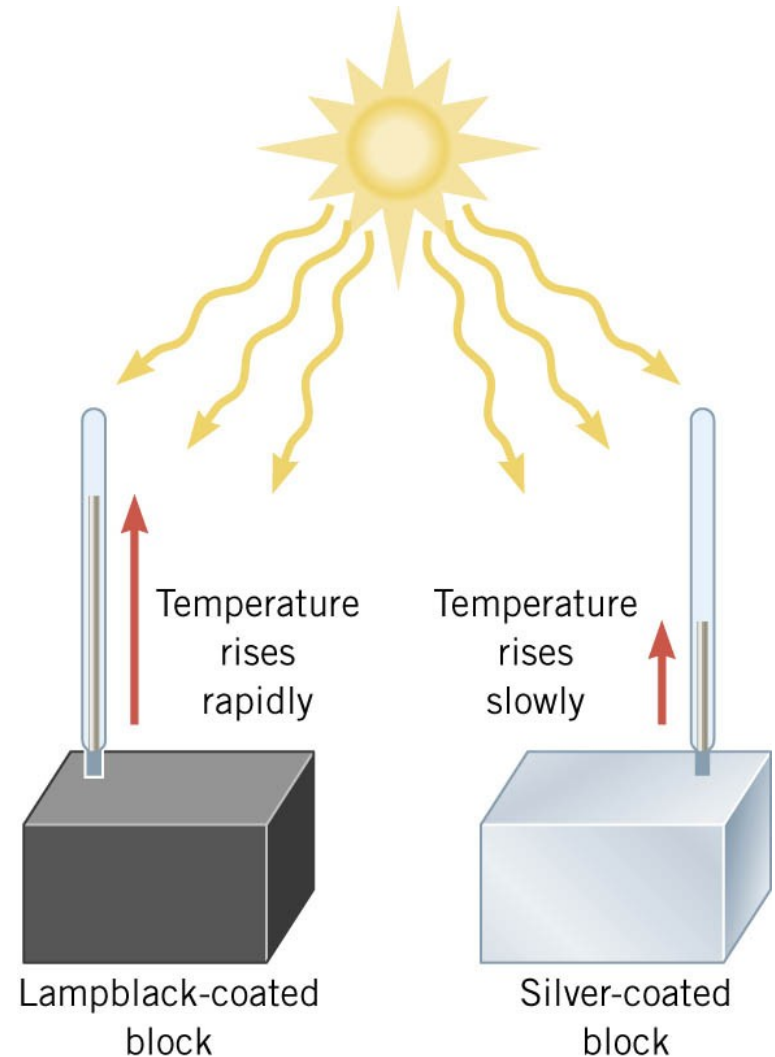
13.3 Radiation

RADIATION

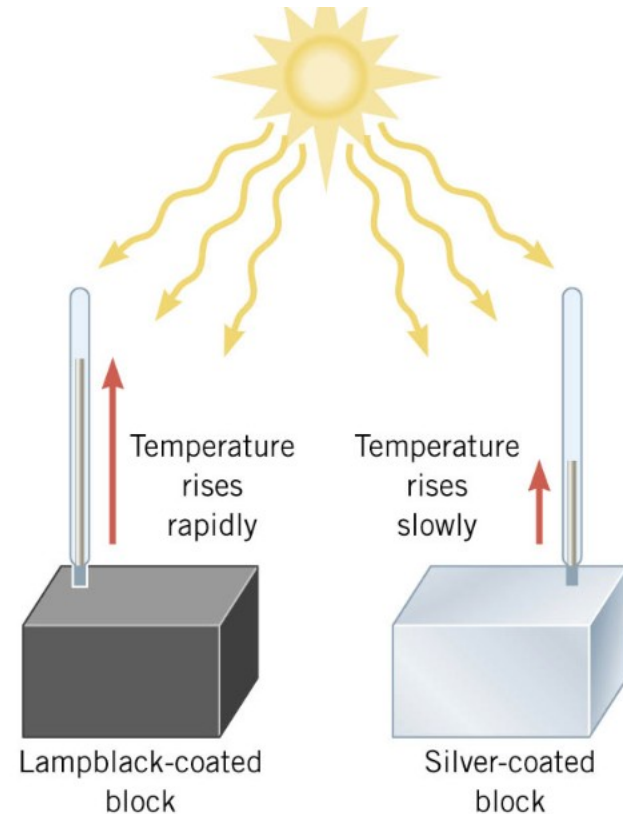
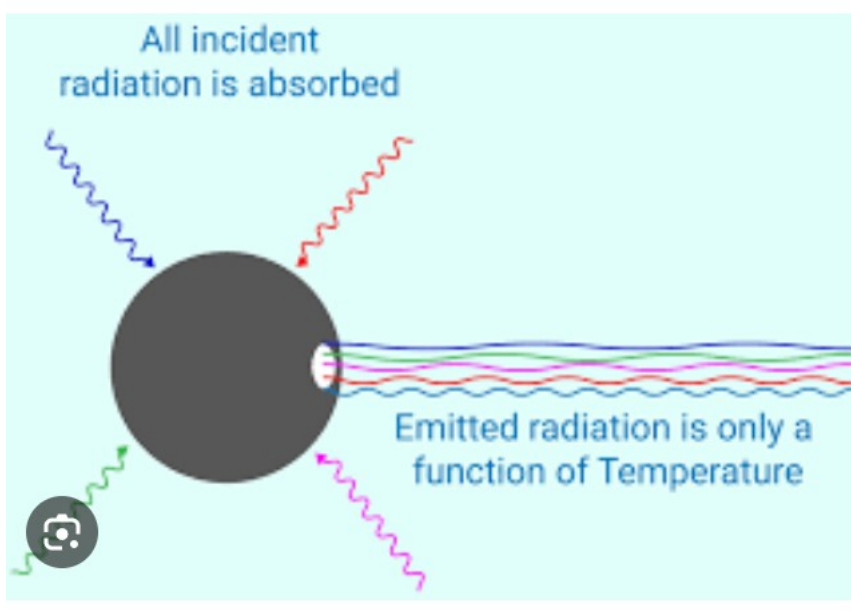
Radiation is the process in which energy is transferred by means of electromagnetic waves.

A material that is a good absorber is also a good emitter.

A material that absorbs completely is called a ***perfect blackbody***.



Black body



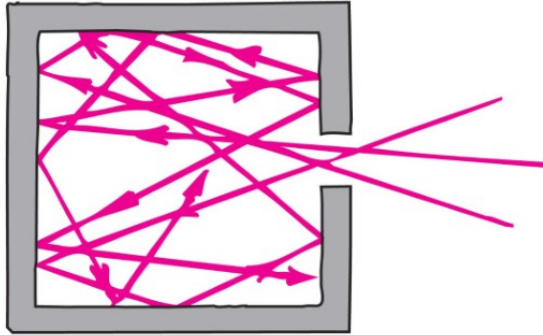
If the boxes are in thermal equilibrium with The outside.

Black body: radiation absorbed = radiation emitted

The white box: most of the light will be emitted.

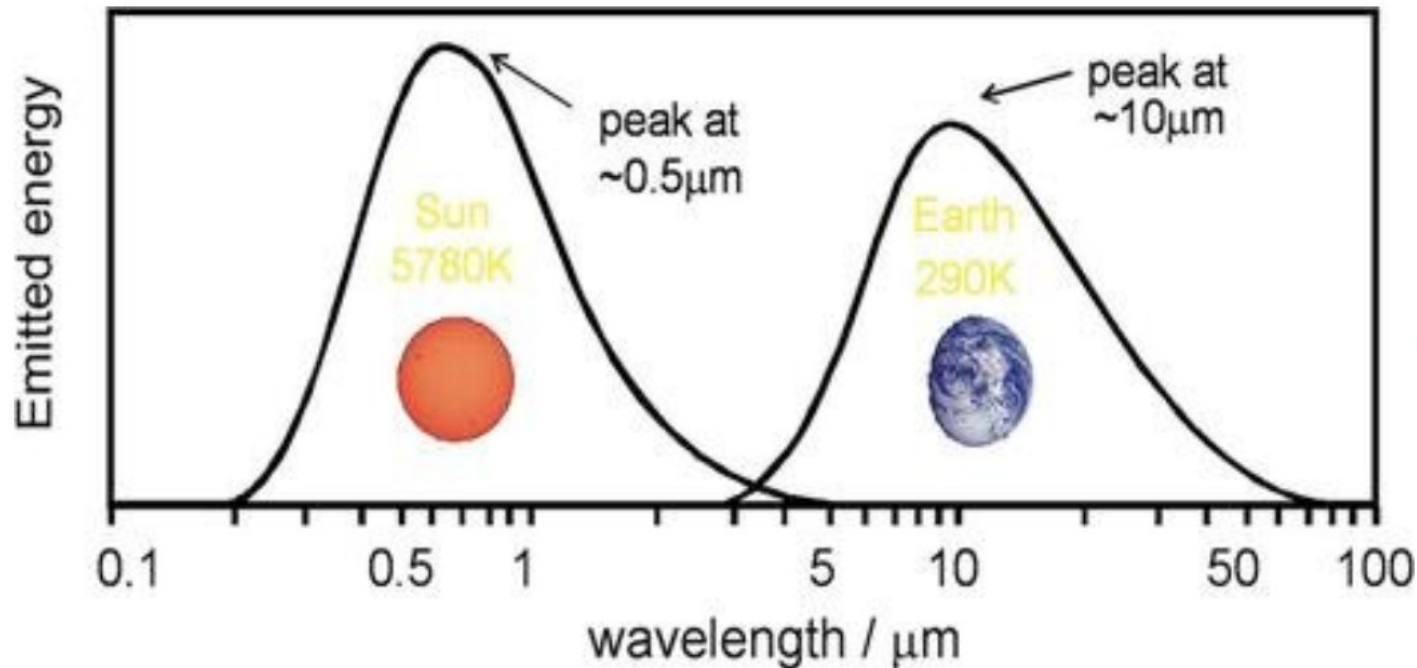
Black body: the radiant energy and the “color” only depends on its temperature.

- Darkness often due to reflection of light back and forth many times partially absorbing with each reflection.



Reference: Paul Hewitt – Conceptual Physics

<https://www.youtube.com/watch?v=xYHapQMDa58>



Black-body emission curves from the sun ($T \sim 5780 \text{ K}$) and the earth ($T \sim 290 \text{ K}$), showing the operation of Wien's Law that $\lambda_{\text{max}} \propto (1/T)$. The two graphs are not to scale.



Wikipedia

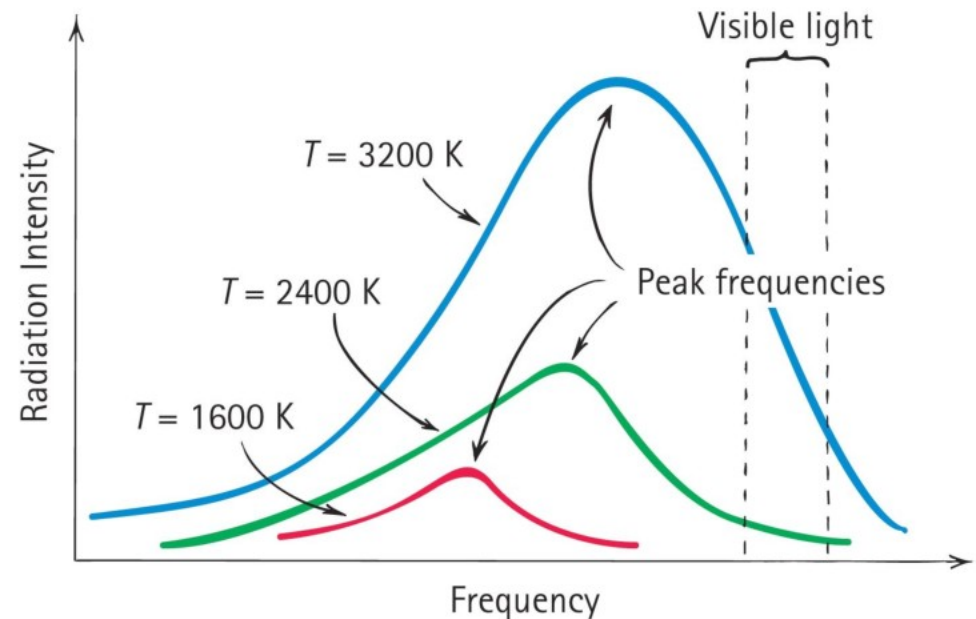
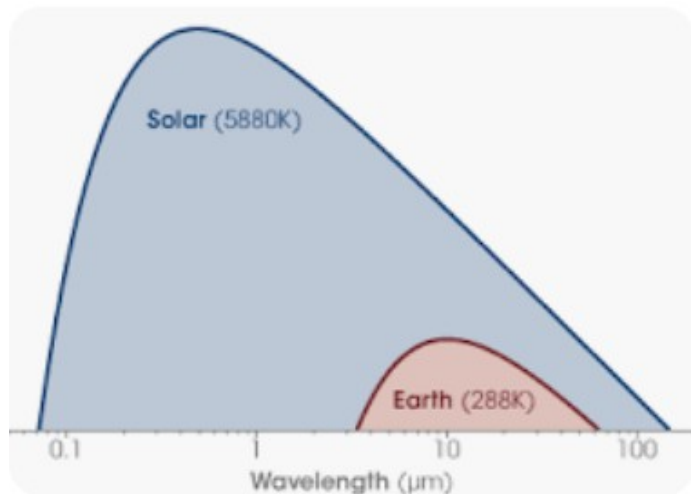
Black-body radiation - Wikipedia

The human body has
A peak wavelength of
About 9500nm (IR)
So 305K temperature

The amount of energy
Per unit time emitted
Or radiant power
Also depends on on the
Temperature
→Max Planck's law black body
radiation law

Black body curve

https://phet.colorado.edu/sims/html/blackbody-spectrum/latest/blackbody-spectrum_all.html



$T > 1000 \text{ K}$ visible light

$T > 1700 \text{ K}$ white hot

The Sun is a $4 \times 10^{26} \text{ W}$ light bulb

13.3 Radiation

The emissivity e is a dimensionless number between zero and one. It is the ratio of what an object radiates to what the object would radiate if it were a perfect emitter.

THE STEFAN-BOLTZMANN LAW OF RADIATION

The radiant energy Q , emitted in a time t by an object that has a Kelvin temperature T , a surface area A , and an emissivity e , is given by

$$Q = e\sigma T^4 At$$

Stefan-Boltzmann constant

$$\sigma = 5.67 \times 10^{-8} \text{ J}/(\text{s} \cdot \text{m}^2 \cdot \text{K}^4)$$

Anatomy of an Equation: Stefan-Boltzmann Radiation Law

Emissivity of surface;
ranges from 0 to 1

Stefan-Boltzmann constant;
 $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$

$$P = eA\sigma T^4$$

Power emitted by a
surface at
temperature T

Surface
area

Stefan-Boltzmann Law

The **Stefan-Boltzmann Law** tells you how much energy per unit area is emitted from a black body.

$$\varepsilon = \sigma T^4 \quad (\sigma = \text{Greek letter "sigma"} = \text{constant})$$



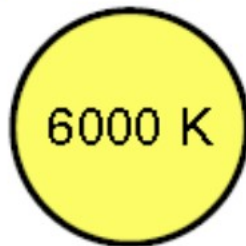
Area in cm^2

Energy per area per sec

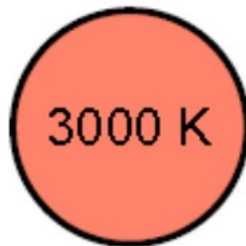
Because of the T^4 power, hotter object emit MUCH more energy per area than cold objects.

Big change in brightness for small change in temperature !

Example: always work with ratios !



6000 K



3000 K

$$\left(\frac{6000}{3000}\right)^4 = 2^4 = 16$$

If a star is 2x as hot, it emits 16 times more power! (if the two stars have the same surface area)

Example 6 A Supergiant Star

The supergiant star Betelgeuse has a surface temperature of about 2900 K and emits a power of approximately $4 \times 10^{30} \text{ W}$.

Assuming that Betelgeuse is a perfect emitter and spherical, find its radius.



13.3 Radiation

$$Q = e \sigma T^4 4 \pi r^2 t$$

$$r = \sqrt{\frac{Q/t}{4 \pi e \sigma T^4}} = \sqrt{\frac{4 \times 10^{30} \text{ W}}{4 \pi (1) [5.67 \times 10^{-8} \text{ J/(s} \cdot \text{m}^2 \cdot \text{K}^4)] (2900 \text{ K})^4}}$$

$$\approx 3 \times 10^{11} \text{ m}$$

A blackbody / thermal equilibrium

In a dynamic situation where the black body is not in equilibrium, its temperature changes according to the net balance of emitted and absorbed radiation:

$$\frac{dE}{dt} = P_{\text{absorbed}} - P_{\text{emitted}}$$



Net emitter



Net absorber

Example 7 An Unused Wood-Burning Stove

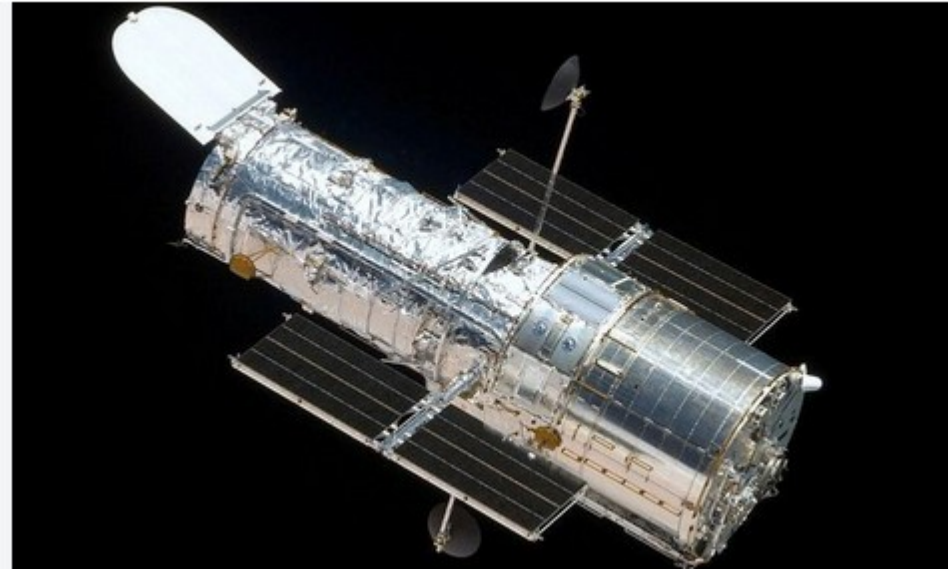
An unused wood-burning stove has a constant temperature of $18\text{ }^{\circ}\text{C}$ (291 K), which is also the temperature of the room in which the stove stands. The stove has an emissivity of 0.900 and a surface area of 3.50 m^2 . What is the *net* radiant power generated by the stove?

Reasoning Power is the change in energy per unit time (Equation 6.10b), or Q/t , which, according to the Stefan–Boltzmann law, is $Q/t = e\sigma T^4 A$ (Equation 13.2). In this problem, however, we need to find the *net* power produced by the stove. The net power is the power the stove emits minus the power the stove absorbs. The power the stove absorbs comes from the walls, ceiling, and floor of the room, all of which emit radiation.

Example 8 The Physics of a Wood-Burning Stove

The wood-burning stove in Example 7 (emissivity = 0.900 and surface area = 3.50 m^2) is being used to heat a room. The fire keeps the stove surface at a constant $198\text{ }^{\circ}\text{C}$ (471 K) and the room at a constant $29\text{ }^{\circ}\text{C}$ (302 K). Determine the *net* radiant power generated by the stove.





A satellite is wrapped in metal foil. The foil is a poor absorber (to protect Electronics from heat when it's Sunside)
And poor emitter. To avoid radiant energy loss.



(a)

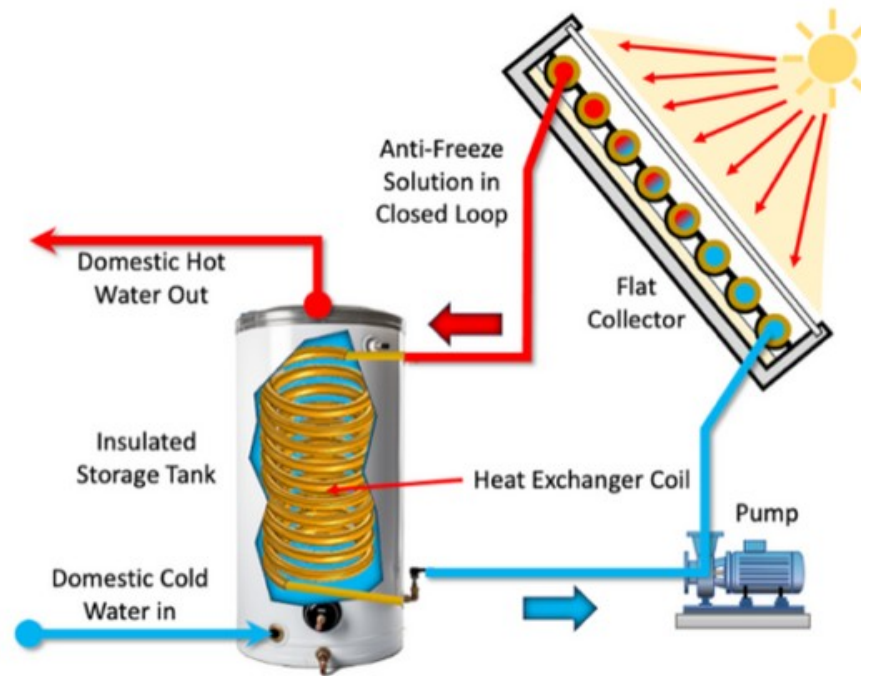


(b)

Figure 13.15 (a) Most lemurs, like this one, are nocturnal and have dark fur. (b) The species of lemur called the white sifaka, however, is active during the day and has white fur. (a. © Wolfgang Kaehler/Corbis; b. © Nigel Dennis/Wildlife Pictures/Peter Arnold, Inc.)

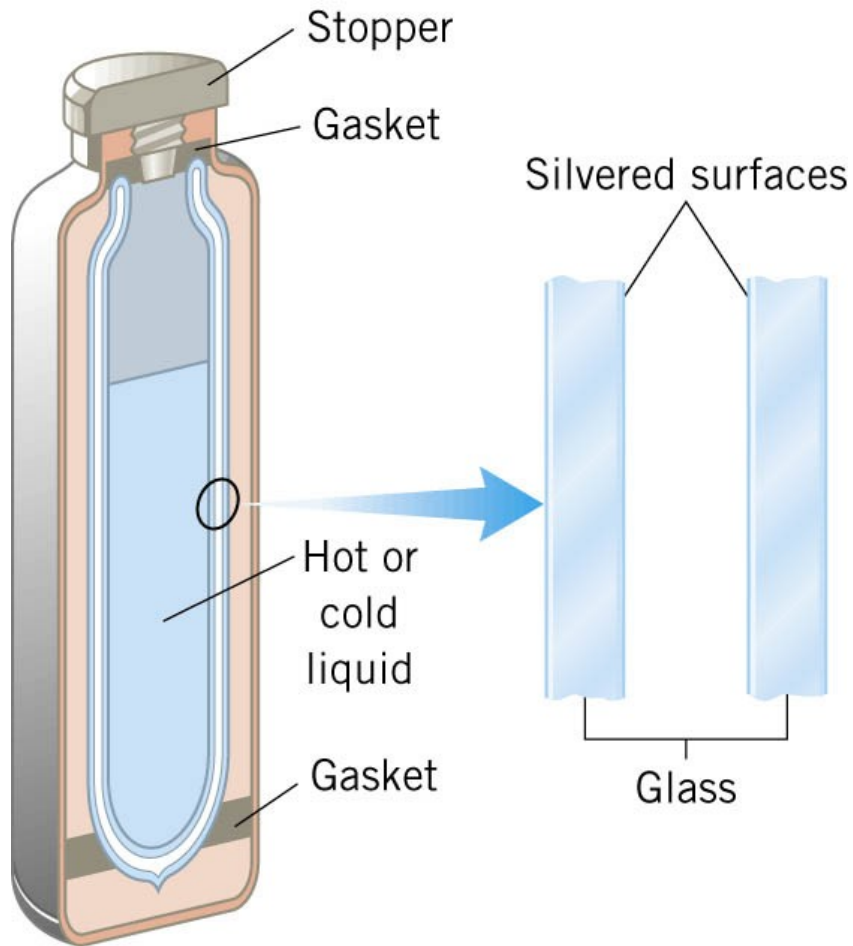


Water solar collector

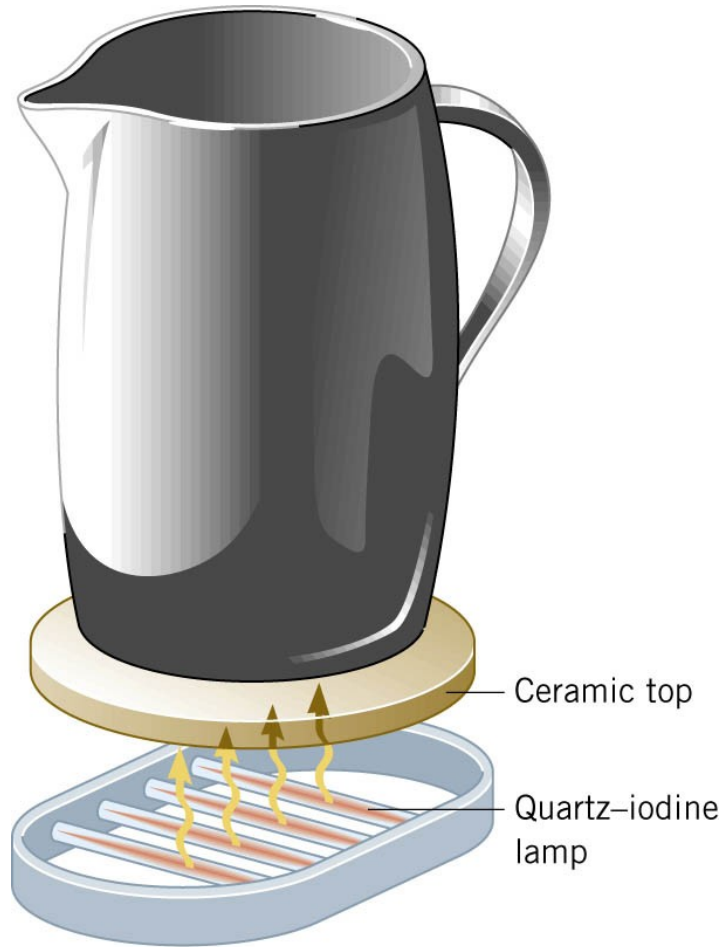


13.4 Applications

A thermos bottle minimizes heat transfer via conduction, convection, and radiation.



13.4 Applications



The halogen cooktop stove creates electromagnetic energy that passes through the ceramic top and is absorbed directly by the bottom of the pot.